

1.) Initial prompt:

a.) 1.) Our hypothesis is this: If patients feel strongly about a topic, they will provide more detailed narratives (richness). Therefore, the topics with the highest volume and strongest sentiments are the priorities that our branding must address to resonate with patients at launch. 2.) The data that I am going to share is Reddit posts concerning the topic of nephrology 3.) Our target audience & goals: people dealing with and finding community in the nephrology realm in reddit, this is who our recommendations are for and what all our data insights are focused on. 4.) So far we've found that the richer the posts the more strongly the person feels about the topic 5.) I'm sharing two visualizations we have made so far as well as the actual data set itself. Act as an expert reviewer specializing in data visualization and audience communication.

2.) Transcript:

1.) Our hypothesis is this: If patients feel strongly about a topic, they will provide more detailed narratives (richness). Therefore, the topics with the highest volume and strongest sentiments are the priorities that our branding must address to resonate with patients at launch. 2.) The data that I am going to share is Reddit posts concerning the topic of nephrology 3.) Our target audience & goals: people dealing with and finding community in the nephrology realm in reddit, this is who our recommendations are for and what all our data insights are focused on 4.) So far we've found that the richer the posts the more strongly the person feels about the topic 5.) I'm sharing two visualizations we have made so far as well as the actual data set itself ...Act as an expert reviewer specializing in data visualization and audience communication

Perfect — this is exactly the kind of work where small changes = massive jump in real-world impact.

I'm going to review this like it's going in front of a healthcare brand team deciding what to say to real patients.

OVERALL ASSESSMENT

You are VERY close to something strong.

Right now:

- Your insight is good

- Your visuals are decent
- But your communication is NOT doing the work for you

👉 The visuals support your hypothesis

👉 But they don't prove it instantly

That's the gap.

VISUALIZATION 1 (Page 1)

What it's trying to show

“Stronger emotion → higher richness”

What actually happens (viewer experience)

- I see boxplots
- I see dots
- I *kind of* see positives are higher

👉 But I have to figure it out myself

Core Problem

- ✗ The insight is not visually obvious
- ✗ There is no anchor (no reference point)
- ✗ No explicit takeaway on the chart

Right now this is:

“Here's some data”

It needs to be:

“Here's the proof of our hypothesis”

EXACT FIXES

1. Add a reference line (THIS IS HUGE)

You NEED:

```
geom_hline(yintercept = avg_richness, linetype = "dashed", color = "red")
```

Label it:

“Average richness”

 Why:

- Instantly shows what is “high” vs “low”
 - Gives executives a benchmark
-

2. Add ONE dominant annotation

Right now: no takeaway

You need:

```
annotate(  
  "text",  
  x = 4,  
  y = 7.5,  
  label = "Higher emotional intensity → richer narratives",  
  fontface = "bold",  
  size = 5  
)
```

 This is the difference between:

- a chart people *look at*
 - vs a chart people *understand*
-

3. Fix color strategy (critical)

Right now:

- Green = positive
- Gray = neutral
- Others unclear

👉 But your INSIGHT is about intensity, not positivity

Do this instead:

- Strong negative → deep red
- Neutral → light gray
- Strong positive → deep green

👉 You want:

emotion strength = visual strength

4. Remove unnecessary noise

- Mixed category = low value → consider removing or de-emphasizing
 - Too many jitter points → reduce alpha more
-

🎯 What this chart SHOULD communicate in 3 seconds:

“Emotionally intense posts are clearly more detailed than neutral ones.”

VISUALIZATION 2 (Top Topics)

What it's trying to show

“Top patient concerns by volume + sentiment”

What actually happens

- I see bars
- I see categories
- I see colors

👉 But I do NOT know:

- what matters most
 - what to prioritize
 - what action to take
-

🚨 Core Problem

- ✗ No prioritization logic
- ✗ No emphasis
- ✗ Everything looks equally important

This is the biggest issue.

🔧 EXACT FIXES

1. Sort by priority (NOT just volume)

You said:

priority = high volume + strong sentiment

👉 So reflect that:

reorder(topic, priority_score)

2. Highlight ONLY what matters

Right now: everything teal

👉 You need contrast:

- ● High volume + negative sentiment → RED (pain points)

-  High volume + positive sentiment → GREEN (opportunities)
 -  Everything else → gray
-

3. Add a clear title that asks a question

Change:

“Treatment Side Effects and Diagnoses Dominate Patient Concerns”

To:

“What Are the Highest Priority Patient Pain Points?”

4. Add direct annotation

Example:

```
annotate(  
  "text",  
  x = 5.5,  
  y = 1,  
  label = "Side effects dominate both volume and emotional intensity",  
  fontface = "bold"  
)
```

 What this chart SHOULD communicate instantly:

“Side effects and diagnoses are the biggest, most emotional patient concerns — this is where branding must focus.”

 **VISUALIZATION 3 (Time Trend)**

What it's trying to show

“Ongoing discussion over time”

What actually happens

- I see lines
 - I see fluctuation
 - I don't know what matters
-



Core Problem

- ✗ No single takeaway
 - ✗ No focus
 - ✗ No interpretation
-



EXACT FIXES

1. Pick ONE message

Based on your data:



Likely insight:

“Discussion remains consistently active over time”

2. Highlight ONE moment

Example:

- Peak month
- Consistent neutral baseline

`annotate("point", ...)`

`annotate("text", label = "Peak discussion period")`

3. Add a strong takeaway label

👉 Example:

“Consistent neutral + emotional discussions indicate ongoing need for support”

🎯 What this SHOULD say instantly:

“This isn’t a temporary issue — patients are consistently seeking support.”

🧠 BIG PICTURE (MOST IMPORTANT)

Right now your work says:

“Here’s what we found”

But it NEEDS to say:

“Here’s what matters — and here’s what to do.”

🔥 FINAL STRATEGIC TAKEAWAY (What YOU should say)

If you present this, say:

Patients express the most detail when they feel the strongest emotions — making emotional intensity the clearest signal of what matters to them.

Side effects and diagnoses dominate both volume and emotional engagement, making them the highest-priority pain points.

Because these conversations persist over time, our branding must consistently address these concerns to build trust and relevance with patients.

If You Want Next-Level

I can:

- Rewrite your exact R code for all 3 visuals
- Help you build a final “client-ready” slide
- Or help you turn this into a storyline that feels like real consulting work

Just tell me 

How is a reference point relevant in visualization 1? Would this be a stronger addition, or would changing the colors of the dots be stronger?

Great question — this is exactly the level of thinking that separates “good-looking” charts from decision-driving ones.

Short Answer

- Reference line = clarity of interpretation
- Color = speed of perception

 If you had to choose one:
The reference line is more important.

Why the Reference Line Matters (More Than You Think)

Right now, your chart shows:

- distributions
- relative differences

But it does NOT answer:

“Is this actually high or low?”

Without a reference point:

The viewer thinks:

“Okay... positives look higher... I think?”

With a reference line:

The viewer instantly sees:

“Anything above this = richer than average → meaningful”

👉 It turns your chart from:

- *relative comparison*
into:
 - absolute interpretation
-

🔥 What This Unlocks Strategically

Your hypothesis is:

stronger emotion → richer narratives

A reference line lets you prove:

emotionally intense posts are not just higher — they are
meaningfully above average

That’s a much stronger claim.

🎨 What About Changing Dot Colors?

Color helps, but it solves a different problem.

Color improves:

- scanning speed
- grouping
- visual appeal

But it does NOT:

- define “high vs low”
- provide a benchmark
- strengthen your argument

👉 So color = helpful

👉 Reference line = interpretation anchor

🏆 **Best Answer: You Should Do BOTH (But Prioritize This Order)**

1. Add reference line (non-negotiable)

This is your foundation

2. Then refine color

This is your amplifier

🎯 **What “Best Version” Looks Like**

Reference line:

- dashed
- labeled “Average richness”
- subtle but visible

Color strategy:

Instead of random/neutral dots:

- Neutrals → light gray
- Strong negative → red
- Strong positive → green

👉 Now your chart communicates:

- Position (above/below average)
- Emotion (color)
- Distribution (boxplot)

That's a 3-layer insight in 1 glance

🔥 One More Expert-Level Insight

If your goal is:

“Higher emotional intensity → richer narratives”

👉 The MOST powerful version would be:

- Keep boxplots
- Add reference line
- AND slightly increase opacity/size of extreme sentiment points

Why?

Because you're visually saying:

“Look here — this is where the story is”

🧠 Final Verdict

Element	Impact Level	Why
---------	-----------------	-----

Reference line	★★★★★	Defines meaning (critical)
Color adjustment	★★★★★	Improves speed/clarity
Both combined	★★★★★	Executive-level clarity

If you want, I can rewrite your exact ggplot code so it:

- adds the reference line correctly
- uses a clean, intentional color system
- and includes a non-overwhelming annotation

That would take your chart from “good” → client-ready.

3. Add a clear title that asks a question Change: “Treatment Side Effects and Diagnoses Dominate Patient Concerns” To: “What Are the Highest Priority Patient Pain Points?” Is this title a relevant change? What are other options for making this chart better?

Great — this is exactly the kind of question that shows you’re thinking like a real analyst, not just formatting charts.

🧠 First: Is the title change actually better?

Your original:

“Treatment Side Effects and Diagnoses Dominate Patient Concerns”

Suggested:

“What Are the Highest Priority Patient Pain Points?”

🔍 Honest answer:

👉 The question title is NOT stronger in your case.

Why the question title is weaker here

A question title works when:

- the chart clearly answers it instantly

Right now, your chart:

- does not clearly show priority
- does not visually highlight pain points

So the viewer thinks:

“Okay... what *are* they then?”

👉 That creates friction.

What you should do instead (this is the real upgrade)

You want a title that:

- states the insight
 - guides interpretation
 - matches what the chart **ACTUALLY** shows
-

Stronger Title Options (Ranked)

Best (Insight + Direction)

“Side Effects and Diagnoses Are the Most Discussed Patient Concerns”

- ✓ Clear
- ✓ Matches data
- ✓ No thinking required

🏆 Even stronger (if you fix highlighting)

“Side Effects Emerge as the Highest-Priority Patient Pain Point”

- ✓ Adds prioritization
- ✓ Slightly more strategic
- ⚠ Only works if you visually emphasize it

🏆 Balanced (if you keep sentiment layering)

“Patient Discussions Are Dominated by Treatment Concerns and Diagnoses”

- ✓ Safe
- ✓ Accurate
- ✓ Slightly less punchy

🎯 Rule to remember (this is big)

- ✗ Question titles = good for exploration
- ✓ Statement titles = better for decision-making

👉 You are in decision-making mode, not exploration.

🔧 Other HIGH-IMPACT Improvements (More Important Than Title)

Honestly — the title is NOT your biggest lever.

These are:

-
1. 🚨 You are not actually showing “priority” yet

You say:

priority = volume + sentiment

But the chart shows:

- mostly volume
- sentiment is visually weak

👉 That's a mismatch.

Fix:

Create a priority signal visually

Option A (simple + effective):

- High volume + negative sentiment →  red
- High volume + positive →  green
- Others → gray

👉 Now “priority” pops out instantly

2.  Everything looks equally important right now

This is your biggest issue.

👉 Executives need:

“Where do I look?”

Fix:

Highlight ONLY top 2–3 topics

Example:

- “Side Effects of Prednisone” → bold color
 - Everything else → muted gray
-

3. 🧠 Add ONE annotation (not many)

Right now: no guidance

You need:

“Side effects dominate both volume and emotional engagement”

Place it near the top bar.

4. 📊 Sort with intention

Right now: looks like volume-based sorting

Better:

- Sort by priority (volume + sentiment)
OR
 - Keep volume, but visually override with color
-

5. 🔥 Reduce cognitive load

Right now:

- multiple colors
- legend
- categories

👉 Too much to process

Fix:

Ask yourself:

“What do I WANT them to see first?”

Then:

- **mute everything else**
-

 **Big Picture Insight (Most Important)**

Right now your chart says:

“Here are the top topics”

But it NEEDS to say:

“These are the problems we must solve first.”

 **Final Recommendation**

Keep a statement title, not a question

 **Best version for you:**

“Side Effects and Diagnoses Are the Most Critical Patient Concerns”

Then fix:

- 1. Highlight priority visually (color)**
 - 2. Emphasize top 1–2 topics**
 - 3. Add one clear annotation**
 - 4. Reduce everything else**
-

If you want, I can:

- Rewrite your exact **ggplot** code to implement all of this
- Or help you create a “this would impress a real client” version

You’re very close — this just needs sharper communication, not new analysis.

For visualization 3, is there any way to keep the messages we have? Do we have to narrow it down to this one message? 1. Pick ONE message Based on your data: 👉 Likely insight: “Discussion remains consistently active over time”

This is a *really* important question — and honestly, this is where people either level up... or stay stuck.

🧠 Short Answer

👉 You don’t have to throw away your messages

👉 But you *DO* have to structure them around ONE primary takeaway

🔥 The Real Principle (This is the key)

A chart can support multiple ideas — but it can only LEAD with one.

Right now, you’re trying to have the chart:

- show persistence over time
- show sentiment breakdown
- show fluctuation
- imply ongoing need

👉 That’s too much for one visual to *lead with*

💣 What Happens If You Keep Multiple Messages

The viewer:

- doesn't know where to look
- doesn't know what matters
- walks away with... nothing clear

👉 This is the worst-case scenario in real-world settings

✅ **The Better Approach (Keep your ideas, structure them)**

Step 1: Choose ONE headline message

Example:

“Patient discussions remain consistently active over time”

Step 2: Let secondary insights SUPPORT it

You can absolutely still include:

- sentiment differences
- fluctuations
- spikes

👉 **BUT they must reinforce the main point**

🎯 **What This Looks Like in Practice**

Your current messages (I'm guessing based on your chart):

- There are ups and downs
- Neutral dominates
- Conversations continue over time

Refactored structure:

🏆 Main takeaway (title or annotation):

“Patient discussions remain consistently active over time”

🥈 Supporting cues in the chart:

- Keep sentiment lines → shows *how* discussion happens
 - Keep variation → shows *when intensity shifts*
 - Highlight one peak → shows *engagement moments*
-

🔧 How to Keep Your Messages WITHOUT Overloading

1. Use hierarchy (this is everything)

Title = main message

Annotation = supporting message

Visual elements = evidence

2. Example of a strong setup

Title:

“Patient Conversations Show Persistent Engagement Over Time”

Annotation (on a peak or mid-period):

“Consistent neutral + emotional discussions indicate ongoing need for support”

👉 Now:

- You kept BOTH ideas
- But one is clearly dominant

3. Use visual emphasis instead of words

Instead of saying everything:

- highlight one peak month
- slightly bold one line (e.g., neutrals)
- fade less important elements

👉 This lets you “say more” without clutter

🧠 Expert-Level Insight (This is big)

If you feel like:

“I have multiple important messages”

👉 That usually means:

You actually have multiple charts trying to live inside one

🏆 Best Options (Ranked)

Option A (Best for clarity)

1 chart = 1 message

→ clean, powerful, executive-ready

Option B (What you’re asking for)

1 main message + 1–2 supporting ideas

→ totally valid, just needs hierarchy

Option C (What to avoid)

3+ equal messages

→ **confusing, forgettable**

Final Answer to Your Question

No — you do NOT have to throw away your messages.

But:

You MUST prioritize one, and make everything else support it.

If You Want Next-Level

I can:

- **Rewrite your exact ggplot code so it keeps your messages but adds hierarchy**
- **Or show you how to split this into 2 complementary charts (which is what real teams often do)**

You're asking the right question — this is exactly how you move from “good student work” to real-world storytelling.

3.) Detailed Analysis

Overall, ChatGPT's response felt focused on specific tweaks to the graph rather than suggestions on how to handle the data as a whole. We were hoping that it would give us advice on how to make the data into more visually appealing graph(s) *overall*, but with the prompt we gave it, it decided to focus on nit-picking certain parts of our visualizations. These critiques were welcome, yet some of them felt like unnecessary “fluff” and were just random changes that ChatGPT gave to us so that we had a higher volume of changes to be made. Essentially, it felt like quantity-over-quality changes were suggested to us. We understand that our initial prompt heavily influenced this, though. We also had to re-run the entire scenario after we forgot to include the context “Act as an expert reviewer specializing in data visualization and audience communication”. Without this context, ChatGPT did not understand that it was

supposed to be genuinely critiquing the graphs, instead of just giving us code to make it look “perfect”. Once we added in the line to act as an expert reviewer, the feedback became more tailored to our scenario. It’s always important to add context!

Something that we did appreciate, though, is that ChatGPT offered to help re-write certain lines of the code to “fix” what was wrong with our graphs in a more *specific* way once we added the context. It’s helpful to have actionable changes that can be made, instead of just random suggestions (such as when it told us to make our title for one of the graphs into a question to gain more attention). After running these graphs through ChatGPT a number of times, though, most of the changes (even in the code) felt unnecessary.